

# Optipeople DemoCNC D-2800M — Operator Manual

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Read this manual before operating the machine. Operators must have completed factory training (see installation manual §10) and be signed off on their competency card.

## 1. Safety summary

| Symbol                                                                                    | Meaning                                 |
|-------------------------------------------------------------------------------------------|-----------------------------------------|
|  DANGER  | Action will cause injury if ignored     |
|  WARNING | Action may cause injury or damage       |
|  NOTE    | Useful information, not safety-critical |

### 1.1 Required PPE

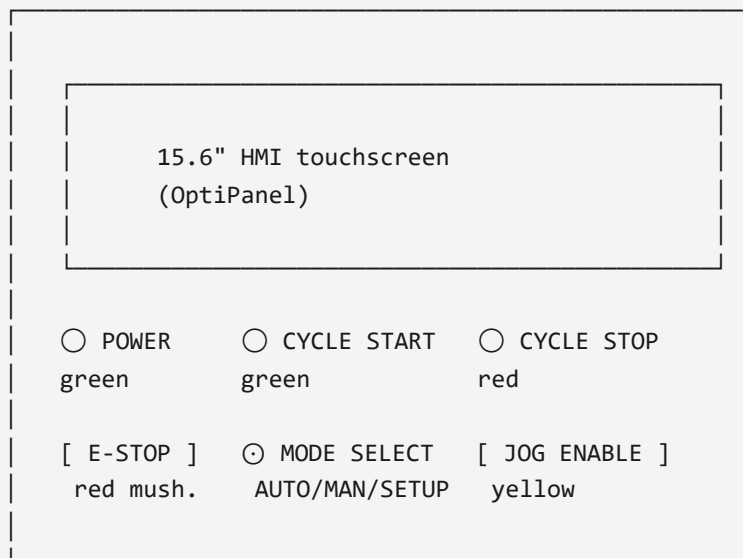
- Safety glasses (EN 166, grade B)
- Hearing protection (EN 352, SNR ≥ 28 dB)
- Cut-resistant gloves **for handling tools and panels only**. Remove gloves before pressing HMI buttons or touching the spindle area.
- Closed-toe safety footwear (S1P minimum)

### 1.2 General rules

1. Never reach inside the safety enclosure while a cycle is running.
2. Never override a door switch. The machine logs all override attempts.
3. Stop the machine with the E-stop before clearing a jammed panel.
4. Do not run the machine with any safety guard removed.
5. The machine produces fine wood dust — wear a P2 filter mask when cleaning out the dust extractor or working on the table.

## 2. Controls

### 2.1 Operator console



USB ports (4× A, 1× C) on the side panel.

Barcode scanner Y-cable plugged into one USB-A by default.

## 2.2 Mode switch

- **AUTO** — runs jobs from the queue. Cycle Start enabled.
- **MAN** — manual jog, manual spindle. Used for setup and troubleshooting. Cycle Start disabled.
- **SETUP** — door interlocks allow opening at reduced speed (max 5 m/min jog, max 1,500 rpm spindle). Requires a held enabling switch on the pendant.

## 3. Daily start-up

1. Visually inspect the machine and surrounding area. Remove offcuts, tools, or anything left on the table.
2. Confirm the dust extractor and vacuum pump are turned on at their own panels. Listen for normal sound.
3. Turn the main isolator Q1 (red handle, cabinet door) to ON.
4. The IPC boots. After ~90 s the OptiPanel HMI loads.
5. Sign in with your operator account. Card-tap login is supported if the optional reader is fitted.
6. The HMI shows the **Homing Required** banner. Press **Home All**. All axes home in this order: Z, X, Y. This takes about 25 s.
7. Open **Status** → **Diagnostics** → **Lubrication** and confirm:
  - Grease reservoir > 25 %
  - Last auto-grease cycle < 8 h ago If either is red, perform the grease procedure (maintenance §3) before running the machine.
8. Run a **15-minute warm-up** if the shop temperature is below 15 °C or if the machine has been idle > 12 h. The HMI offers this on the start screen and cycles each spindle from 30 % to 100 %.

## 4. Loading a panel

1. Open the front door of the work zone you intend to use.
2. Place the panel against the rear positioning bar. The bar rises automatically when you tap **Load Zone A** or **Load Zone B**.

3. Slide the panel left until it contacts the X-side reference stop.
4. Close the door. Press **Confirm Loaded** on the HMI.
5. The HMI commands the movable beam to descend, the top pressers to extend, and the rear bar to retract. You'll hear a sequence of clicks over ~3 s.
6. **i** Min panel size 250 × 200 mm. Smaller pieces will not be held reliably — use the small-part jig (PN: OP-JIG-SP01).

## 5. Running a job

### 5.1 By barcode (preferred)

1. Scan the panel's barcode with the handheld scanner.
2. The HMI loads the job, shows a 3D preview, and highlights the machining operations.
3. Press **Cycle Start**.
4. Watch the first cuts through the safety window. If anything looks wrong (excessive vibration, smoke, wrong tool depth), press the nearest E-stop and call a supervisor.

### 5.2 By manual selection

1. Tap **Jobs** → **Browse**.
2. Filter by customer, date, or part name.
3. Select a job. The preview opens.
4. Confirm material thickness — the auto-thickness probe will verify when you press Cycle Start, but if you know the panel is non-standard you can pre-set it here.
5. Press **Cycle Start**.

### 5.3 During the cycle

- The HMI shows operation progress, current tool, and estimated remaining time.
- The opposite zone's door is unlocked — you can load the next panel while the current one is being machined. This is the main productivity feature.
- **Do not** open the active zone's door. The machine will halt with alarm **E-2010 Door open during cycle**.

### 5.4 End of cycle

1. The machine lifts the head, retracts pressers, and beeps.
2. The HMI shows **Cycle Complete**.
3. Open the door, remove the panel.
4. Clean offcuts and chips off the vacuum table with the supplied brush. Do **not** use compressed air on the table — it drives dust into the vacuum gaskets and shortens their life.

## 6. Setup mode operations

Setup mode (key-switch + held enabling button) is used to manually position the head, test a single operation, or measure a tool. Door interlocks are eased — opening the door drops speed but does not stop the machine.

### 6.1 Manual jog

1. Switch to MAN mode.

2. Press JOG ENABLE.
3. Select the axis (X / Y / Z) on the HMI.
4. Use the jog wheel. Default increment is 0.1 mm — change with the x1 / x10 / x100 buttons.

## 6.2 Set tool length (vertical setter)

1. Load the tool in S1 (use ATC if it's already in the magazine).
2. From the HMI, **Tools** → **Calibrate** → **Length (Vertical)**.
3. The head moves over the laser setter and runs the calibration.
4. The new length is shown — confirm and save.

## 6.3 Set tool length (horizontal probes)

1. Load tool in S3 or S4.
2. **Tools** → **Calibrate** → **Length (Horizontal S3/S4)**.
3. The head moves to the touch probe and dabs it twice.
4. Save when complete.

## 7. Loading new tools into the ATC

1. Switch to SETUP mode.
2. **Tools** → **Magazine** → **Load Slot N**.
3. The magazine indexes; slot N is exposed at the front opening.
4. ⚠️ Wait for the green "READY" LED on the slot. The pull-stud solenoid is only released then.
5. Insert the tool holder, push until it clicks. The slot LED turns blue.
6. On the HMI, enter the tool data: type, diameter, length (or use calibrate from §6.2), feeds, and life counter target.
7. Repeat for additional slots.

## 8. End-of-shift shutdown

1. Finish the current cycle. Do not abort mid-cycle just to leave on time — partial parts must be removed from the table and the vacuum left running long enough to keep the panel held.
2. **HMI** → **Shutdown** → **Park**. The head moves to the parking position (X = 5,500, Y = 80, Z = 200).
3. Wait for the HMI to display **Safe to Power Off**.
4. Turn isolator Q1 to OFF.
5. Close the air supply ball valve V1 if leaving for the weekend or longer.
6. ⓘ Leave the dust extractor running for 60 s after shutdown to clear the ducting. Most modern collectors have a built-in run-on timer.

## 9. Quick troubleshooting (operator level)

| Symptom                 | Try first                                              | If that fails                                                                       |
|-------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------|
| Panel slips during cut  | Check vacuum gasket strip, brush table clean           | Replace gasket — maintenance \$6.2                                                  |
| Burn marks on edge      | Check tool wear; reduce feed by 10 %                   | Replace tool, recalibrate length                                                    |
| Chatter / vibration     | Confirm tool is properly seated and ER32 nut torque    | Stop and call supervisor — could be spindle bearing                                 |
| Dust escaping enclosure | Check Ø 200 dust hose clamps                           | Inspect filter on extractor — likely clogged                                        |
| HMI frozen              | Tap <b>Status</b> → <b>Lock screen</b> → <b>Unlock</b> | Restart HMI from <b>Shutdown</b> → <b>Restart HMI</b> (machine stays in safe state) |
| Will not home           | Check no objects on table; ensure all doors closed     | See error codes E-1xxx (errors doc)                                                 |

For any alarm code (E-xxxx, W-xxxx), look it up in document [06-error-codes.md](#).

## 10. Cleanliness habits

Operators are expected to keep the machine in good condition:

- Every cycle: brush offcuts off the table.
- Every 2 hours: open the dust drawer (front-left, push-to-release) and empty if more than 1/2 full.
- End of shift: wipe the HMI screen with the microfibre cloth.
- End of shift: clear chips from the ATC slots — chips packed into the taper cause the next tool change to fault.
- End of week: vacuum the cabinet intake filter grille (cabinet door, bottom).

## 11. When in doubt — STOP

The right behaviour for an operator who is unsure is always to stop and ask. The machine, the parts, and your hands are all worth less than the job. A two-minute supervisor call beats a damaged spindle every time.